

Evaluating Missing Data Handling Techniques Under Different Missingness Mechanisms

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Introduction

Missing data is one of the most pervasive challenges in epidemiological, clinical, and social science research. Whether due to participant dropout, non response, data corruption, or logistical limitations, incomplete datasets are the norm rather than the exception. If not properly addressed, missing data can lead to biased estimates, inflated standard errors, and ultimately invalid inferences^[1]. As public health decisions increasingly rely on large-scale observational data, the need for robust and transparent missing data handling has become critically important. Rubin's (1976)[2] framework offers the foundation for understanding missing data mechanisms: Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR). Under MCAR, the probability of missingness is unrelated to both observed and unobserved data, meaning that analyses restricted to complete cases remain unbiased. However, in practice, data are rarely MCAR. When data are MAR, i.e., the probability of missingness depends only on observed data, more sophisticated modeling approaches are needed to recover unbiased estimates. MNAR presents the greatest challenge: when missingness is related to the unobserved value itself, statistical adjustment becomes far more complex and often requires strong, unverifiable assumptions[3].

In applied research, the default strategy is often complete-case analysis due to its simplicity and ease of implementation. Yet under MAR or MNAR conditions, this can result in substantial bias and loss of efficiency[4]. Single imputation methods, such as mean or median substitution, are equally problematic, as they distort the distribution of the data and underestimate uncertainty. In contrast, Multiple Imputation by Chained Equations (MICE) and other model-based techniques offer a principled solution by simulating the distribution of missing values multiple times and pooling results to account for variability due to imputation[5]. Despite their growing popularity, these methods are not always correctly applied, and their performance can vary depending on the nature and extent of the missingness[6]. Simulation studies provide a framework for evaluating the relative performance of these missing data techniques. By generating data under controlled conditions and imposing different missingness mechanisms, we can compare the accuracy and reliability of each approach across a range of outcomes, such as bias in regression coefficients, confidence interval coverage, prediction accuracy, and model fit indices like AIC or BIC[7; 8]. These evaluations are essential for identifying which imputation strategies are most appropriate under real-world constraints.

Relevance to Obesity Research

Obesity presents a growing public health challenge worldwide, contributing to increased risk of cardiovascular disease, type 2 diabetes, hypertension, musculoskeletal disorders, and certain cancers. According to the World Health Organization (2024)[9], global obesity prevalence has tripled since 1975, and over 650 million adults are currently classified as obese. In Latin America, where countries like Mexico, Peru, and Colombia are experiencing rapid nutrition and lifestyle transitions, obesity rates have surged across all age groups[10; 11; 12]. The determinants of obesity are multifactorial, including sociodemographic variables, poor dietary

patterns, inadequate physical activity, sedentary behaviors, and harmful habits such as smoking and alcohol consumption[13;14]. These risk factors are often collected via self-report or web-based tools, making them especially vulnerable to missingness due to social desirability bias, recall error, or issues with technology access. Understanding how missing data may bias the estimation of such relationships is critical for valid epidemiologic inference.

Study Objective

The objective of this project is to conduct a simulation study using the [UCI Obesity](#) dataset to assess how various imputation techniques perform under MCAR, MAR, and MNAR scenarios. We will compare complete-case analysis, single imputation (mean or median), and MICE across performance metrics including bias, confidence interval coverage, model fit, and predictive accuracy. In doing so, this study aims to provide empirical guidance for method selection in missing data analysis and highlight the importance of aligning the imputation strategy with the underlying missingness mechanism.

Data Availabiliy

All codes and datasets for this project can be found at our [GitHub Repository](#).

References

1. Rubin, D. B. (1987). *Statistical analysis with missing data*. Wiley.